



MATES Computer Science

Senior Capstone Project Bi-Weekly Progress Report

Project Title	Olive the World
Team Members	Zachary Ciappa , Ella Jacinto , and Sabrina Shi
Dates Covered by Report	March 3, 2026 - March 20, 2026
Link to Github	https://github.com/jackandtoes/Olive-the-World

1. Summary of Project

Olive the World! is a web game consisting of a variety of minigames. Players will travel around the world as Olive the olive with the goal of becoming the best chef on the planet. Each minigame is modeled after an existing popular game; the Philippines game resembles Crossy Road, Italy is inspired by Pipe Mania, and Mexico is modeled after Fruit Ninja. Olive the World! will be free for public use, opening minds to diverse cuisines from around the world.

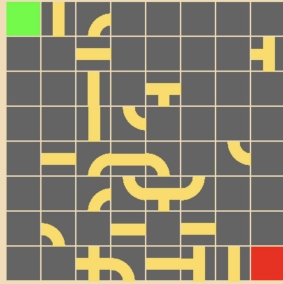
2. Summary of Progress this Period

During this progress period, we began personalizing the games, tailoring them to our countries. For the Italy and Philippines games, we both set up a way to have multiple levels. We began replacing our placeholder graphics in favor of personalized icons. The Mexico game was updated to include special prizes and some cool visual changes. The Philippines game was updated to fix tiling issues and other problems with moving. The Italy game was updated to include two complete levels with a win condition. Additionally, there are now limits on the amount of moves you can make on the Italy game. We have been keeping up with the Github repository and making commits fairly frequently.

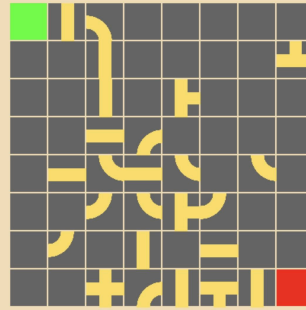
3. Detailed Progress this Period, separated by Team Member Zac's Accomplishments

I have continued the 7 hour CodeCademy course on Phaser game development, and I am currently about 40% done with it. I have focused my energy on creating my puzzle game. Players now have max amounts of moves they can make before the pastas are "locked" and can not be rotated. One thing I added that I think really adds to the experience is randomizing the pasta orientation when you begin the level.

Clicks:40

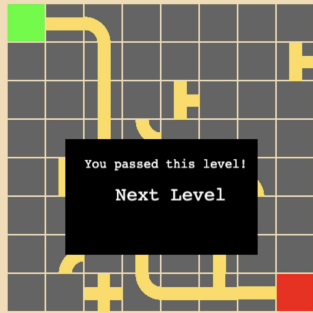


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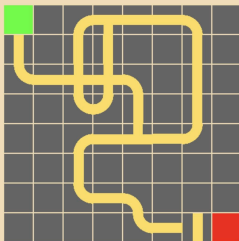
The biggest hurdle I got over during this progress period was creating a win condition for my game. Now, a win screen will pop up when you have all the pasta in the correct place.

YOU WIN!



In addition to this, I also created a new second level for the pasta. This level is tougher and requires more moves to be made to win. The win condition took a long time to create and figure out. I had to do everything by hand in order to ensure that the win condition only popped up once someone got the exact order and rotation indices of the pasta.

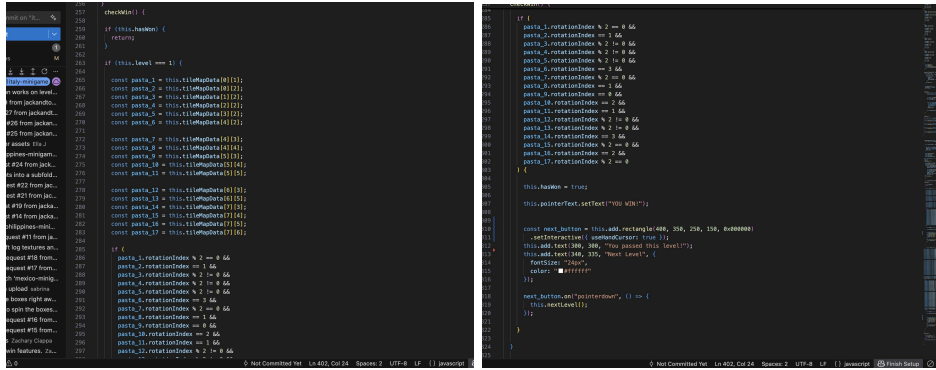
Clicks: 14



YOU WIN!



Here are some screenshots of the win condition code for level 1:



Ella's Accomplishments

I am continuing the 7 hour CodeCademy course on Game Development using Phaser and I am currently 81% done with it. I completed the full Phaser Animations lesson which was very helpful, and I now know how to add detailed walking animations to the player which I hope to maybe implement on the map game selection scene. Additionally, the Phaser Animations lesson went over multiscreened games in more depth, which was also helpful for when I start working on the code for the currency and shop system. The final CodeCademy course on Visual Effects has you code a 2D platformer game, which will be very helpful to use as a guide when creating the extra country games (Brazil & Egypt).

I have fixed the tiling issue that I had previously where the olive would go on the edge of the log and make the game look weird, however that has opened up a new can of worms where the location tracking system will sometimes get messed up so item collection won't work and the only way to fix it is to reset the player position by going on a log and getting off of the log and trying again. I tried playtesting the game many times to figure out the exact circumstances that cause this issue, but to no avail. I also was able to implement a final victory screen for when you complete all 5 levels, which I was able to reach after playtesting my game a bunch of times when trying to fix the player location issue. Additionally as you can see from the screenshot, I switched out the brown rectangle log texture to a better looking log prototype, so I had to slightly adapt my code to that change. I used textures I found online, but I am planning on actually making our own log design.



```
showVictory() {
  const width = this.scale.width;
  const height = this.scale.height;
  const bg = this.add.rectangle(width/2, height/2, 500, 400, 0x000000, 0.9);

  this.add.text(width/2, height/2 - 120, "Victory!", {
    fontSize: "56px",
    fill: "#ff0000"
  }).setOrigin(0.5);
  this.add.text(width/2, height/2 - 40, "All Ingredients collected!", {
    fontSize: "28px",
    fill: "white"
  }).setOrigin(0.5);
  this.add.text(width/2, height/2 + 10, "You completed all 5 levels!", {
    fontSize: "24px",
    fill: "#00ff00"
  }).setOrigin(0.5);

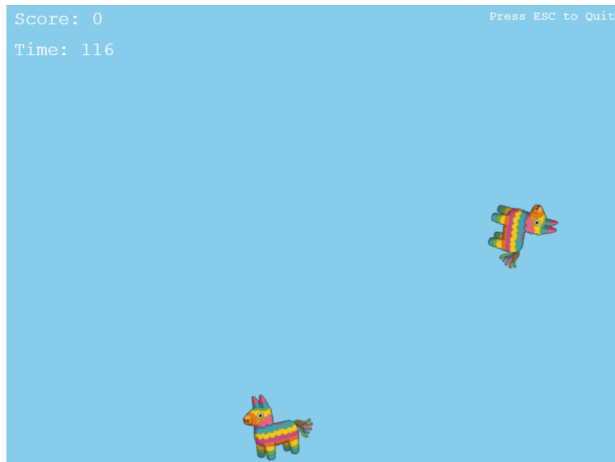
  const playAgainBtn = this.add.text(width/2, height/2 + 80, "Play Again", {
    fontSize: "28px",
    fill: "#00ff00"
  }).setOrigin(0.5).setInteractive();
  playAgainBtn.on("pointerdown", () => {
    this.scene.restart({ level: 1 });
  });
}
```

Sabrina's Accomplishments

In addition to continuing the CodeCademy course, I have also begun researching how to animate to make our game more engaging and created some sprite sheets to use later on.

In my Mexico minigame, I added code to make the game progressively harder as the timer counts down, spawning more targets and more chili bombs to get in the way. I have also added golden pinatas that spawn rarely towards the end of the game to allow the player to increase their scores quickly. Additionally, I have begun working on the aesthetics of the game. Using Gemini, I generated images for the pinatas, golden pinatas, and chilies. These replaced the boxes that were originally used as place holders during the last sprint.

With 116 seconds left:



With 22 seconds left (includes the three targets):



To make the physics of the game look more realistic, I added code to spin the targets at a random angle as they're thrown in the air.

```
const chileData = {
  el: chile,
  startX,
  endX,
  startY,
  peakY,
  startTime: Date.now(),
  duration: 3000,
  clicked: false,
  angle: Phaser.Math.Between(-3, 3)
};
```

```
for (let i = this.chiles.length - 1; i >= 0; i--) {
  const chileData = this.chiles[i];

  if (chileData.clicked) continue;

  chileData.el.angle += chileData.angle;

  const elapsed = now - chileData.startTime;
  const t = elapsed / chileData.duration;
```

I also changed how many points each target added or subtracted. The normal pinatas added 10 points and the golden pinatas added 50 points. The chilies would subtract 30 points from the total score. To make it easier for the players to see their score, the amount of points then won or lost shows up when the target is clicked.



Says "+10" where I hit the pinata then the number fades away

Lastly, I added a win and lose condition that pops up once the 2 minutes of the game is up. The menu gives the player the choice to restart the game or to return back to the map. However, this game still needs to be play tested to figure out a good passing score.

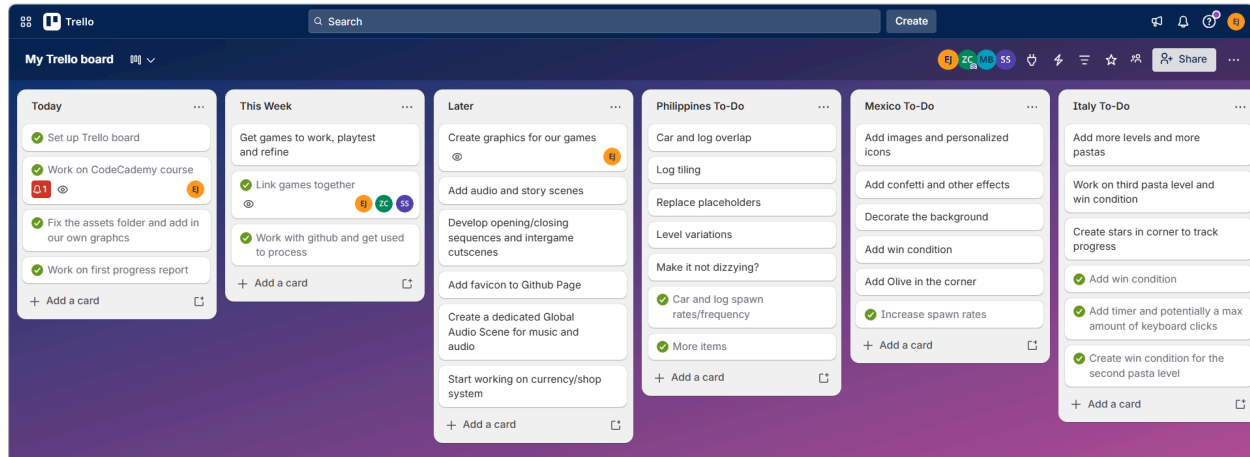


4. Difficulties Encountered this Progress Period

Ella accidentally ran something in her terminal and it caused her whole philippines-branch to revert all her changes so it undid a significant amount of progress. Luckily she had a couple snippets of code in another Google Doc saved, but she still had to fix a lot of it.

While uploading the images for the Mexico game, the images seemed to either be blurry or slightly flicker when playing the minigame. This is likely due to the dimensions of the image being too large or too small, so they will have to be resized and reuploaded later on.

5. Updated Trello Board and Discussion



6. Tasks to Be Worked on in Next Progress Period

Zac:

During this next project period, I will be working on expanding my game and the number of levels. I want to finish the three levels that I have left. However, this may be tricky because each level takes a long time to add pasta and create a win condition. I also want to create a game over screen. Right now, when you run out of moves to make, the pastas just can not be rotated. I plan to add a screen where you can reset your puzzle. This will take you back to the first puzzle and destroy your progress. Right now, I have green and red tiles as the beginning and end areas. I want to change this so they are pots of sauce (the beginning) and a plate (the end). Finally, I want to add our little olive character somewhere in here. It will likely be added to the side and look like it's watching you play.

Ella:

I still need to fix the car and log overlap visual glitch.



Additionally, when I made the grid invisible, the game started to make me really dizzy whenever I looked at it. I think I still have to find the correct balance for the speeds of the car and log, where it's not going too fast as to make you dizzy but it's not too slow where playing it is really boring. I also want to work on more distinctions between the levels instead of just having different ingredients and collection numbers. I also am currently working on different map layouts for each level (ex: level 1 is mainly grass rows but level 5 is mainly water rows). I also want to work on better versions of my placeholders. Finally, I would like to get started with the universal currency integration and shop feature, which will be easier to implement with the knowledge gained from the Phaser Animations CodeCademy lesson.

Sabrina:

For the next sprint, I want to add in animations of candy exploding every time a pinata is clicked on. I will also need to start playtesting the game to refine the win condition to be fair but also challenging. If time permits, I want to incorporate the Olive character somewhere on the screen. I was planning on adding Olive by the scoreboard and having reactions every time a golden pinata or a chili is hit. I also want to fix the image glitching issue and decorate the background of the game more.

In addition to progressing the Mexico minigame, I want to start cleaning up the start screen and the map screen. Right now, they mostly consist of placeholders but I want to make them more aesthetically pleasing and add in some animations in the background.

7. Additional Information

So far, it seems we are progressing quickly and will likely have time to add more minigames later on.